Use-case flow: request emergency blood

Blood bank inventory & emergency donor matcher

Team12:
PES1UG24AM128(KARANAM SRUJANA)

Primary actor	Emergency requester
Supporting actors	Blood bank system, SMS gateway, Donor
Preconditions	Emergency requester is authenticated on the system. The blood bank inventory ledger and SMS gateway are online and reachable.
Postconditions	Compatible donors within the search radius have been notified and the notification is logged, or the requester has been informed that no match was found.

Main success scenario

1 Emergency requester selects “Request emergency blood” and enters blood type, quantity, and delivery location.

2 System validates the input for completeness and correct blood-type format.

3 System cross-references the request against real-time blood bank inventory (use case: Check inventory).

4 If sufficient stock exists, system reserves the matching units and proceeds to fulfillment.

5 If stock is insufficient, system identifies eligible donors of a compatible blood type within a 10 km radius (use case: Notify donors).

6 System sends SMS alerts to all matching donors via the SMS gateway.

7 System logs the notification event with timestamp, donor list, and blood type requested.

8 System displays a confirmation to the requester showing request status and estimated response time.

9 Use case ends successfully.

Alternate flow

A. No compatible donors respond within the alert window

- a1. System monitors donor responses against a configured timeout (e.g., 10 minutes).

- a2. If no donor confirms availability within the window, system automatically escalates the request to nearby partner blood banks within a wider radius.
- a3. If escalation also yields no match, system notifies the requester that no compatible donor or stock was found and logs the failed request for manual follow-up by the blood bank manager.

Related use cases: this flow includes **Check inventory** and **Notify donors** («include»), and interacts with **Track expiry alerts** («extend» of Manage blood stock) indirectly, since expired units are excluded from the inventory .